

Congyi Nai

congyi.nai@worc.ox.ac.uk | Google Scholar

Education

University of Oxford

PhD student, Intelligent Earth Programme

Sep 2026 – Present

Oxford, UK

University of California, Los Angeles (UCLA)

Ph.D. studies, Atmospheric and Oceanic Sciences

Sep 2025 – Apr 2026

Los Angeles, CA, USA

Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences

M.S. in Physical Geography

Sep 2021 – Jun 2024

Beijing, China

Tsinghua University

B.Eng. in Hydraulic Engineering

Sep 2017 – Jun 2021

Beijing, China

Professional Experience

Institute of Atmospheric Physics, Chinese Academy of Sciences

Research Assistant

Jul 2024 – 2026

Beijing, China

- Research in generative AI for weather forecasting, data assimilation, precipitation nowcasting.
- Developed probabilistic deep-learning systems for ensemble prediction and uncertainty quantification.

Publications

Selected and updated from the author's publication record. #Equal contribution.

Selected Publications

1. **Nai, C.**, Pan, B., Liang, Y., & Chen, X. (2026). "Cast3: Translating numerical weather prediction principles into data-driven forecasting." *arXiv:2605.01599*.
2. **Nai, C.**, Chen, X., Yang, S., Liang, Y., Xiao, Z., & Pan, B. (2025). "Boosting weather forecast via generative superensemble." *npj Climate and Atmospheric Science*, 8, 377.
3. Yang, S.#, **Nai, C.#**, Liu, X., Li, W., Chao, J., Wang, J., et al. (2025). "Generative assimilation and prediction for weather and climate." *arXiv:2503.03038*.
4. **Nai, C.**, Pan, B., Chen, X., Tang, Q., Ni, G., Duan, Q., Lu, B., Xiao, Z., & Liu, X. (2024). "Reliable precipitation nowcasting using probabilistic diffusion models." *Environmental Research Letters*, 19, 034039. **Editor's Choice**.
5. **Nai, C.**, Liu, X., Tang, Q., Liu, L., Sun, S., & Gaffney, P. P. J. (2023). "A novel strategy for automatic selection of cross-basin data for training machine learning-based runoff models." *Water Resources Research*.
6. Ou, Z.#, **Nai, C.#**, Pan, B., Zheng, Y., Shen, C., Jiang, P., Liu, X., Tang, Q., Li, W., & Pan, M. (2025). "Probabilistic diffusion models advance extreme flood forecasting." *Geophysical Research Letters*, 52(15), e2025GL115705.
7. Sun, S., **Nai, C.**, Pan, B., Li, W., Li, L., Li, X., Foufoula-Georgiou, E., & Lin, Y. (2026). "Fusion of multi-source precipitation records via coordinate-based generative models." *Nature Communications*, 17.

Skills

Programming Python; PyTorch (fluent); scientific computing; deep learning

Research Generative modeling; diffusion models; weather and climate prediction; extreme weather prediction

English TOEFL: Reading 24, Listening 23, Speaking 20, Writing 20